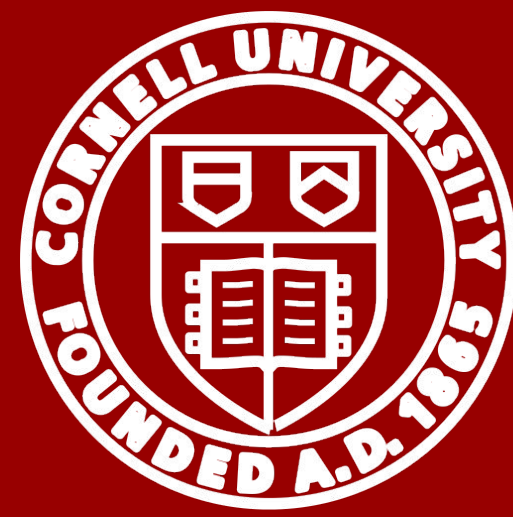


Understanding the Cost of Compression: BERT vs. DistilBERT

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Introduction

Motivation: Large pretrained models like BERT achieve high performance but carry prohibitive **computational and memory costs** for resource-constrained settings.

The DistilBERT Solution: A compressed version of BERT trained via **knowledge distillation to be smaller and faster** while retaining core language understanding.

Performance Claims: The original paper reports DistilBERT is **40%** smaller, **60%** faster, and retains **~97%** of BERT's performance.

Research Goal: Quantify the specific "cost of compression" by **identifying** where **accuracy drops** in downstream **classification**.

Hypothesis: DistilBERT's loss is concentrated in tasks requiring **compositional logic** (negation) and **long-range dependencies**.

Methodology

We compare model predictions on individual examples to characterize qualitative failure modes, such as negation and long-context reasoning.

- Datasets:** IMDb, SST-2, MRPC
- Models:** BERT-base and DistilBERT. DistilBERT is a distilled student model that preserves the encoder-only Transformer design of BERT while using fewer layers.
- Tasks:** Downstream sentiment classification on SST-2 and IMDb.
- Evaluation:** Accuracy, F1, training time, inference time, and parameter count.
- Analysis:** Prediction-level outputs, confidence scores, and text-length statistics are used to study where DistilBERT loses performance relative to BERT.

Results

DistilBERT is smaller and usually faster than BERT, while keeping similar performance. It is slightly worse on SST-2 and IMDb, but slightly better on MRPC.

Paper claim	Our measurement	Trend
40% smaller	39% smaller (67.0M / 109.5M params)	✓
60% faster (training)	30% (SST-2) / 48% (IMDb) / 40% (MRPC) faster	Partial
Retains ~97% of capability	96.9% (SST-2) / 99.1% (IMDb) / 101.8% (MRPC) of BERT's accuracy	✓ +

	SST-2 (872 val)		IMDb (25,000 test)		MRPC (408 val)	
	BERT	DistiBERT	BERT	DistiBERT	BERT	DistiBERT
Accuracy	0.928	0.899	0.924	0.916	0.821	0.836
F1	0.930	0.902	0.925	0.916	0.876	0.883
Accuracy gap (pp)		-2.87		-0.84		+1.47
Training time	391 s	275 s	1013 s	530 s	52 s	31 s
Training speed-up		1.42x		1.91x		1.65x
Parameters	109.5M	67.0 M	109.5 M	67.0 M	109.5 M	67.0 M
Inference latency (ms / sample)	2.24	1.98	3.48	1.87	7.62	8.31
Inference speed-up		1.13x		1.86x		0.92x
Agreement rate		95.3%		95.2%		86.3%

Discussion

- Systematic Performance Gaps**
- Task-Dependent Accuracy:** Gap is wider on short sentences (SST-2: **-2.87 pp**) than long documents (IMDb: **-0.84 pp**).
 - The "MRPC Paradox":** DistilBERT outperforms BERT (**+1.47 pp**); reduced capacity prevents over-fitting to lexical overlap.
 - Inconsistent Speed-up:** Training is consistently faster (**1.4x–1.9x**), but less benefit on short inputs (IMDb: **1.86x** vs. SST-2: **1.13x**).
 - Altered Boundaries:** Agreement drops from **~95%** (sentiment) to **86.3%** (MRPC), proving distillation changes complex relational logic.

Where DistilBERT Fails

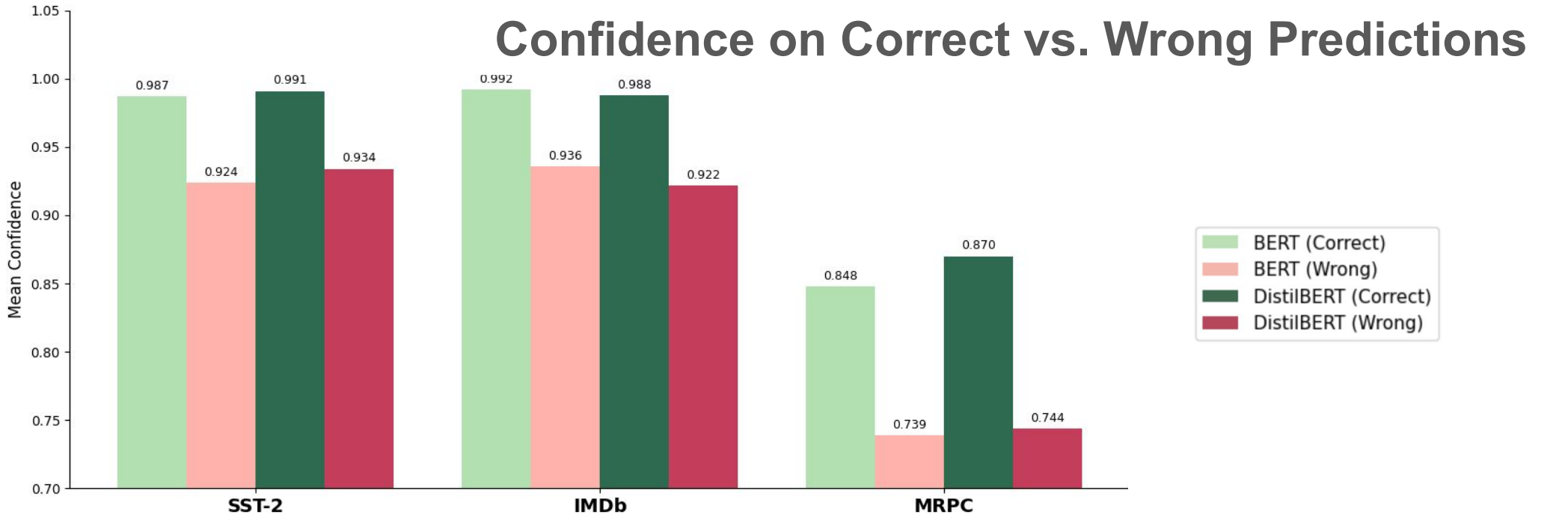
Linguistic Complexity	Long Doc Reasoning	Confidence
when sentiment depends on structure: - Negation ("don't like") - Contrast ("good but bad") - Sarcasm / idioms → relies on surface words instead of structure	long reviews e.g. IMDB - mixed sentiment - delayed conclusions - opinion shifts → overweights early signals → misses final judgment	Overconfident when wrong → poor uncertainty estimation

Key Insights

- Distillation ≠ performance loss; → changes decision boundary
- DistilBERT sometimes wins (contrast cases)
- Outperforms BERT on MRPC → less overfitting
- Acts as regularization in low-data settings

I didn't think it was going to be this good."

BERT: positive! 😊
DistilBERT: negative!



Conclusion

- Validation of Claims:** Confirmed DistilBERT retains **97–99% accuracy** while providing **1.4x–1.9x faster training**.
- Contextual Loss:** Compression costs are not uniform; performance drops significantly on tasks requiring **compositional logic** or **long-range dependencies**.
- The MRPC Paradox:** Reduced capacity acts as a **regularizer** in low-data settings, allowing the student to outperform the teacher.
- Deployment Risks:** High accuracy belies **poor calibration**; DistilBERT is "confidently wrong," making it unsuitable for safety-critical tasks.

Future Work

- Robustness Distillation:** Incorporate **contrastive learning** or **saliency matching** to better capture negation and sarcasm.
- "In-the-Wild" Analysis:** Compare BERT-DistilBERT gaps on **noisy, non-benchmark** data.
- Uncertainty Calibration:** Apply **Scaling or Smoothing** techniques during distillation.
- Architecture Comparison:** Investigate whether long-context reasoning failures persist **across other "shallow" models**.

References

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Our results replicated the paper’s trends. To study the cost of distillation and how it works, we focus on the performance gap between BERT and DistilBERT. We compare the performance of BERT-base (teacher) and DistilBERT-base (student) on SST-2 and IMDb. On SST-2, BERT outperforms DistilBERT by 2.87%, with disagreements on 41 out of 872 validation examples (~4.7%). Among these, BERT is correct in 33 cases, while DistilBERT is correct in 8. On IMDb, the gap is smaller but still consistent: BERT achieves 0.84% higher accuracy. The two models disagree on 1190 out of 25000 reviews (~4.8%), with 700 BERT-only-correct and 490 DistilBERT-only-correct cases.

The performance gap is **systematic** rather than random, driven by specific linguistic and structural challenges.

- **DistilBERT struggles when sentiment depends on structure rather than individual words.** For example, negation (e.g., “don’t think I liked it”), contrast (“good but... bad”), and sarcasm and idioms. It relies more on surface-level lexical cues and often fails to model scope and interaction.
- **DistilBERT performs worse on longer reviews containing mixed sentiment (summary + opinion), delayed conclusions (final sentence matters most), and temporal shifts in opinion.** It tends to over-weight early or frequent signals and misses final judgments.
- **DistilBERT is often overconfident in incorrect predictions**, indicating poor uncertainty estimation.

However, DistilBERT is not uniformly worse. In some contrast-heavy cases (e.g., “bad but I liked it”), it can outperform BERT, suggesting that **distillation changes decision boundaries rather than simply degrading performance**. Moreover, on MRPC, the larger model’s extra capacity becomes a liability, leading to over-confidence in "lexical overlap" (assuming two sentences are paraphrases just because they share words). DistilBERT’s more "conservative" decision boundary allows it to win by refusing to call these pairs paraphrases, suggesting that **distillation acts as a form of regularization in low-data settings**.